SOWJANYA SURADA

Computer Science Engineering Graduate — IT Fresher

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate (B.Tech, 2026) with a strong foundation in programming, data structures & algorithms, databases, and software development. Hands-on project experience in Computer Vision and Machine Learning — including a self-driving car simulation built end-to-end with a working autonomous demo — and currently expanding into full-stack development. Quick learner with three internships across AI/ML and Python development, adaptable to software engineering, data, or analyst roles.

TECHNICAL SKILLS

- Languages: Java ,Python,C, C++
- Core CS: Data Structures & Algorithms, DBMS, OOP fundamentals
- Frameworks/Libraries: TensorFlow, Keras, YOLOv5, YOLOv11n, OpenCV, MediaPipe, PyTorch, Scikit-learn, CNN, NumPy, Pandas
- Tools & Platforms: Streamlit, Flask, SocketIO, Tkinter, Git, MySQL, Jupyter, Google Colab, MS Excel, PowerPoint

ACADEMIC PROJECT — FINAL YEAR

Self-Driving Car Simulation using Deep Learning | *B.Tech Capstone, Team of 4* Dec 2025 – Apr 2026

- Built an end-to-end autonomous driving pipeline using the Udacity Self-Driving Car Simulator, applying behavioral cloning to map camera images directly to steering angles
- Designed and trained an NVIDIA-style CNN (TensorFlow/Keras) on preprocessed and augmented driving data (cropping, resizing, normalization, flipping) to predict continuous steering commands
- Built the real-time inference pipeline with Flask + SocketIO to stream simulator camera frames to the trained model and return steering/throttle commands
- Achieved successful autonomous driving on straight roads and moderate curves in the simulator with no human intervention

PERSONAL PROJECTS

Drone Detection using YOLOv11n | *Personal Project*

- Built a real-time object detection model for drones using annotated Roboflow datasets, achieving 85%+ accuracy while staying lightweight for real-time use
- GitHub: github.com/sowjanya-py/drone-detection

Plant Disease Detection App | *Personal Project*

- Trained a YOLOv5 model on a Roboflow plant-disease dataset; built a Streamlit UI showing bounding boxes, confidence scores, and treatment suggestions
- GitHub: github.com/sowjanya-py/plant-disease-detection | Live demo available

AIVA (Telegram Bot) · Face Landmark Detection · Python GUI Tools | *Personal Projects*

- AI assistant bot for conversational task automation; real-time facial landmark detection using MediaPipe; Tkinter utilities and a web-based student report card generator

EXPERIENCE

AI & Computer Vision Intern | *AIMER Society* May – Jul 2025

- Developed CNN and YOLO-based models for image classification and detection tasks
- Built a sign language detector and multiple image classifiers

DSA Intern | *CodeBeat* Jun – Aug 2024

- Solved data structures and algorithms problems, strengthening core problem-solving skills

Python Development Intern | *Technohacks EduTech* May – Jun 2024

- Built Python automation tools and GUI applications using Tkinter

EDUCATION

B.Tech, Computer Science Engineering — CGPA: 8.0 | *Miracle Educational Society Group of Institutions, JNTU-GV* 2022 – 2026

Senior Secondary (XII), CBSE — 8.0 | *Kendriya Vidyalaya, INS Kalinga, Visakhapatnam* 2022

Secondary (X), CBSE — CGPA: 7.9 | *Kendriya Vidyalaya, INS Kalinga, Visakhapatnam* 2020

CERTIFICATIONS & ACTIVITIES

- Data Science with Python — Teachnook (in association with IIT Bhubaneswar)
- Currently pursuing Java Full Stack Development training — Core Java, Spring Boot, Hibernate, REST APIs, MySQL, React (Naresh IT, Hyderabad, 2026)
- Tech Fest Lead; Hackathon participant | Languages: Telugu, Hindi, English